

Kinetic Isotope Effects

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Abstract

The effect of isotopic substitution on the chemical equilibrium and kinetic rates of reactions. Vibrational changes caused by the change in atomic mass result in small but measurable changes in ΔG that are observed in changes in equilibrium constants or rate constants. These perturbations can be used to bring further understanding to chemical reaction mechanisms.

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1 Structural Changes Change Energy Surface of Reaction

When we change the structure of a molecule involved in a particular reaction we change the shape of the energy surface of the reaction (of which the reaction coordinate is a part). We can change the electron density at the reaction center, the steric environment around the reaction center or both. By changing the reaction energy surface we change the rate of reaction and, sometimes, the mechanism of the reaction.

1.1 Substitution Reactions

Figure 1 shows the energy surface defined by the bond distances from reaction center to nucleophile and leaving group in a substitution reaction. Depending on changes to the surface the reaction rates will change and the mechanism will be either SN1 or SN2 depending on the relative energies of the corresponding transition states. In the diagram below, the SN1 and SN2 transition states are very close in energy.

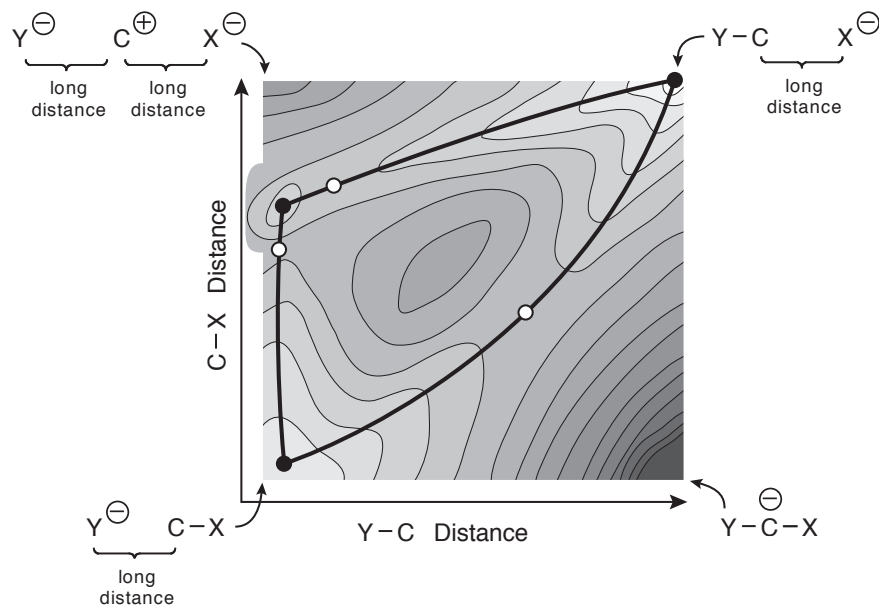


Fig. 1: Example Reaction Surface for Substitution Reactions.

1.2 Elimination Reactions

Figure 2 shows a reaction energy surface for an elimination reaction. Structural changes in the reactant will alter the energies of the transition states and will change reaction rates and possibly the mechanism.

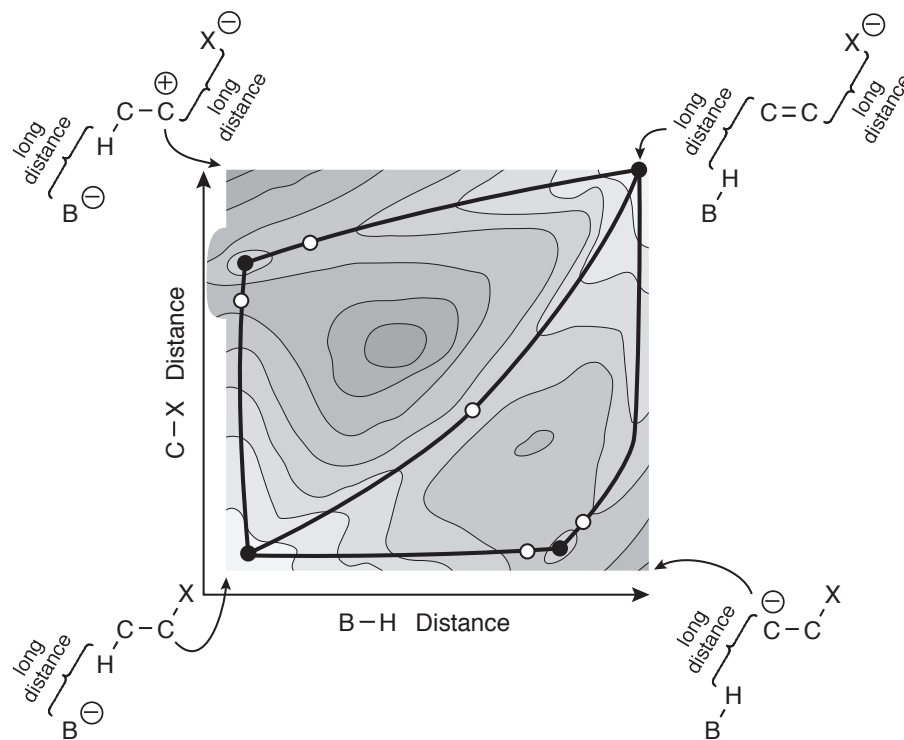


Fig. 2: Example Reaction Surface for Elimination Reactions.

1.3 Isotope Changes Do Not Alter The Energy Surface of a Reaction

The energy of a reaction surface is not affected by changing an atom to a different isotope of that atom. The size of an atom is determined by its electron cloud. The energy of its bond is determined by the interaction of its atomic orbitals with the atomic orbitals of another atom determine the bond strength. The valence shell electrons are influenced by the charge of the nucleus and the shielding of the non-bonding inner-shell electrons. Adding a neutron to the nucleus will not alter the reaction surface.

1.3.1 Isotopic Substitutions Affect Rates and Equilibria

After stating that isotopic substitutions have no effect on the reaction energy surface, why is it that isotopic substitutions at the reaction center often alter the equilibrium constant or rate constant for a reaction. Table 1 on the following page shows some examples of the following types of isotope effects.

Equilibrium Isotope Effect: When a substitution for a heavier isotope results in a lower equilibrium constant. e.g., exchanging the proton involved in an acid dissociation for deuterium alters the acid equilibrium for a phenol. The equilibrium constant is smaller with the deuterium substitution. The ratio of the equilibrium constants for the acid equilibrium of dinitrophenol and pyridine in benzene is $K_H/K_D = 1.40$.

Inverse Equilibrium Isotope Effect: When a substitution for a heavier isotope results in a higher equilibrium constant. e.g. Exchanging the proton involved in addition of an oxime to an aldehyde. The equilibrium constant is larger with the deuterium substitution. The ratio of the equilibrium constants for the addition of hydroxylamine to benzaldehyde is $K_H/K_D = 0.74$.

Kinetic Isotope Effect: When a substitution for a heavier isotope results in a lower rate constant for the reaction. e.g. exchanging the protons on the electron-poor molecule in a Diels-Alder Condensation results in a change in the rate of reaction. In the case of cyclopentadiene with deuterated maleic anhydride the rate is slightly faster than in the unsubstituted case. $k_H/k_D = 0.95$. (observe that $k_H/k_D < 1$; this is an inverse kinetic isotope effect.)

We have so far considered only the effect of deuterium/hydrogen substitutions. If a heavier atom is being substituted by an isotope, it is termed a heavy atom isotope effect. Substituting ^{37}Cl for ^{35}Cl in the substitution of basic thiophenol for chloride in pentenyl chloride results in a kinetic isotope effect of $k_{35\text{S}}/k_{37\text{S}} = 1.0089$.

Solvent Kinetic Isotope Effect: exchanging the protons on a protic solvent with deuterium will result in a change in the rate of reaction involving proton-solvent transfers or general base/general acid catalysis by solvent. For the acid equilibrium between 2-acetohexanone and acetate in water (or deuterated water) we see that $K_H/K_D = 1.20$.

1.4 Isotope Effects Can Infer Mechanism

Depending on the reaction mechanism we will have different types of kinetic isotope effects. We will only be considering proton/deuterium substitutions from now on.

Tab. 1: The Isotope Effect In Chemical Reactions

	Reaction (L=location of isotopic substitution)	Isotope Effect
(a)		$\frac{K_H}{K_D} = 1.40$
(b)		$\frac{K_H}{K_D} = 0.74$
(c)		$\frac{k_H}{k_D} = 0.95$
(d)		$\frac{k_{35S}}{k_{37S}} = 1.0089$
(e)		$\frac{k_H}{k_D} = 1.03$

Primary Kinetic Isotope Effect: When the bond that connects the proton/deuteron involved is broken during the rate-determining step of a reaction. Primary kinetic isotope effects are large (significantly smaller or larger than 1.0). For the reaction below $K_H/K_D = 4.6$

Secondary Kinetic Isotope Effect: When the proton/deuteron is attached to the reaction center but the bond is not involved in the reaction mechanism. Secondary kinetic isotope effects are small (slightly larger or smaller than 1.0). For the reaction below $K_H/K_D = 1.03$.

2 Kinetic Isotope Effect and Vibrational Energy Levels

2.1 Introduction

The equilibrium constant, K_{eq} , is related to change in energy, ΔE . The relationship between the relative changes in energy between the starting materials

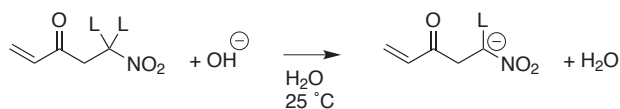
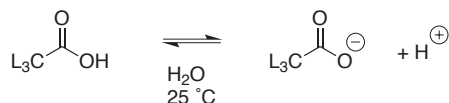
(a) Primary Kinetic Isotope Effect $K_H/K_D = 4.6$ (b) Secondary Kinetic Isotope Effect $K_H/K_D = 1.03$

Fig. 3: Types of kinetic isotope effects

and products or between the starting materials and transitions state for normal and deuterated reactions is described below.

The equilibrium constant for a reaction is given by...

$$\Delta G = -R \cdot T \cdot \ln K_{eq} \quad (1)$$

Where ΔG is the change in free energy of the molecule between initial and final states. The ΔG value is taken to be the same the ΔH value for this equation as we are assuming that any changes in entropy (ΔS) are very small (more accurately we are assuming that any ΔS for the deuterated system is identical to any ΔS in the non-deuterated system and thus they cancel out leaving only the changes in ΔH between the two systems). And, of course, since no work is being done (we are assuming the volume does not change) we can say that the ΔH term is equal to the changes in potential energy of the molecule itself during the reaction, reflected by ΔE . So we can use the difference in energy levels between the initial and final states of the reaction as the total free energy difference. We are saying that $\Delta E = \Delta G$.

The electronic energy, E_e , is the same for both deuterated and non-deuterated systems since isotopic substitution does not change the electron energies of a molecule. So the ΔE_e for the reaction is the same for both deuterated and non-deuterated systems. Thus the change in ΔE_e between a deuterated and non-deuterated system is zero ($\Delta \Delta E_e = 0$)

The vibrational energy, E_v , will be different for deuterated and non-deuterated systems and the ΔE_v for the deuterated reaction will be different than the ΔE_v for the non-deuterated reaction. Thus the change in ΔE_v between a deuterated and non-deuterated system is the difference between these two values ($\Delta \Delta E_v = \Delta E_{v(H)} - \Delta E_{v(D)}$)

The rotational energy, E_R , and translational energy, E_T , of the molecules are too small to be significant contributors to the energy differences between deuterated and non-deuterated systems. So any changes in the ΔE of a reaction when

deuterated and non-deuterated is due to the different ΔE_v values that may result with isotopic substitution.

The total ΔE for a reaction is given in Equation 2.

$$\Delta E = \Delta E_e + \Delta E_v + \Delta E_R + \Delta E_T \quad (2)$$

We will ignore $\Delta E_R + \Delta E_T$ due to their very small size relative to ΔE_v . So we are saying that the total ΔE for a reaction can be simplified to Equation 3.

$$\Delta E = \Delta E_e + \Delta E_v \quad (3)$$

For the non deuterated reaction we have a change in energy between initial and final states of $\Delta E_{(H)} = \Delta E_{e(H)} + \Delta E_{v(H)}$. For now we will deal with just the total, $\Delta E_{(H)}$. Similarly, for the deuterated reaction we have a change in energy between initial and final states of $\Delta E_{(D)} = \Delta E_{e(D)} + \Delta E_{v(D)}$. For now we will deal with just the total, $\Delta E_{(D)}$.

Using the above two equations, and knowing that $\Delta E_{e(H)} = \Delta E_{e(D)}$ we see that...

$$\begin{aligned} \Delta\Delta E &= \Delta E_{(H)} - \Delta E_{(D)} \\ \Delta\Delta E &= (\Delta E_{e(H)} + \Delta E_{v(H)}) - (\Delta E_{e(D)} + \Delta E_{v(D)}) \\ \Delta\Delta E &= \Delta E_{v(H)} - \Delta E_{v(D)} \end{aligned} \quad (4)$$

$\Delta\Delta E$ is the difference in the difference between initial and final molecules between deuterated and non-deuterated systems. $\Delta\Delta E$ relates to the isotopic effect described by k_H/k_D (kinetic) and K_H/K_D (equilibrium) by...

$$\frac{K_{eq(H)}}{K_{eq(D)}} = e^{\frac{-\Delta\Delta E}{RT}} \quad (5)$$

So any significant changes in the equilibrium or reaction rate will be due to changes in the vibrational energy levels due to isotopic substitution.

Note: In this equation $\Delta\Delta E = \Delta E_{v(H)} - \Delta E_{v(D)}$ (from Equation 4). In Equation 9 below we use $-\Delta\Delta E = \Delta E_{v(D)} - \Delta E_{v(H)}$, which is the exact same equation and so the minus sign does not appear in Equation 9.

2.1.1 Summary

- Electronic energy did not change with isotopic substitution.
- Force constants for the vibrations of importance in the initial molecule or the final molecule (be it product of an equilibrium or transition state structure) are not changed by isotopic substitution.

- Vibrational energy levels are changed by isotopic substitution due to changes in the reduced mass term for vibration frequency.
- The change in equilibrium constant or rate constant is directly related to change in the change in energy difference between initial molecule and final molecule (be it product of an equilibrium or transition state structure) being considered.

2.1.2 How does the isotope effect relate to vibrational energy levels?

The isotope effect is defined as the ratio between the equilibrium constants or rate constants of a reaction.

For isotope equilibrium effects the isotope effect is given by the ratio of the equilibrium constants $\frac{K_{eq(H)}}{K_{eq(D)}}$.

By rearranging Equation 1 we can see that

$$K_{eq} = e^{\frac{-\Delta G}{R \cdot T}} \quad (6)$$

So for the ratio of the equilibrium constants we see

$$\frac{K_{eq(H)}}{K_{eq(D)}} = \frac{e^{\frac{-\Delta G_{(H)}}{R \cdot T}}}{e^{\frac{-\Delta G_{(D)}}{R \cdot T}}} \quad (7)$$

Which can be simplified to

$$\frac{K_{eq(H)}}{K_{eq(D)}} = e^{\frac{\Delta G_{(D)} - \Delta G_{(H)}}{R \cdot T}} \quad (8)$$

And since we are considering that $\Delta G = \Delta E$ we have

$$\frac{K_{eq(H)}}{K_{eq(D)}} = e^{\frac{\Delta E_{(D)} - \Delta E_{(H)}}{R \cdot T}} \quad (9)$$

3 Relationship Between Changes In The Force Constant And Changes In $\Delta\Delta E_v$ Due To Isotopic Substitution.

As have seen, the change in the change in energy between initial and final molecules being considered is related to changes in the vibrational energy levels in the molecules due to isotopic substitution. Here I will hopefully review the subject for you and present a more rigorous mathematical treatment of the issue.

We will be dealing with energy differences from the ground state vibrational level ($V = 0$) of the initial molecule to the ground state vibrational level ($V = 0$) of the final molecule (be it product of an equilibrium or transition state

structure) being considered. The energy of the ground state vibrational level can be calculated from its frequency of vibration as related by...

$$E_v = \left(V + \frac{1}{2}\right) \cdot h \cdot v \quad (10)$$

Where h is the Planck constant, $h = 6.626 \times 10^{-34} \text{ J s}^{-1}$ and v is the frequency (which can often be obtained from spectroscopic data) in s^{-1} .

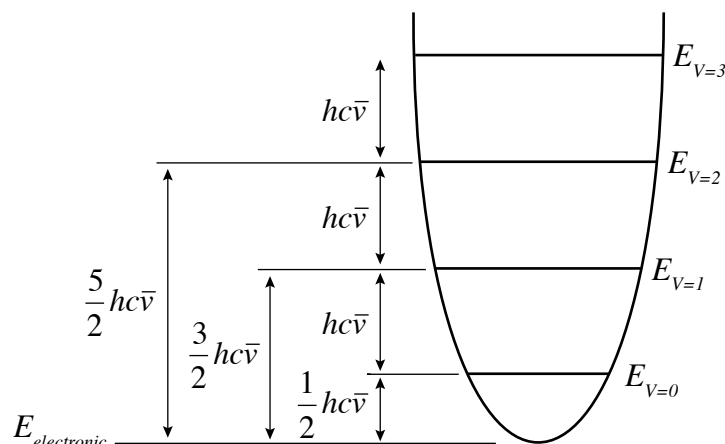


Fig. 4: The idealized model of a harmonic oscillator for C–H stretching vibration.

Assuming that the spectroscopic value of v was for the peak represented by the transition from $V = 0$ to $V = 1$ (A most likely situation is, at room temperature, the majority of the molecules will be in the vibrational ground state) we can use that v to calculate the vibrational energy at $V = 0$. (see Figure 3 on page 8).

(Aside: in a perfect harmonic oscillator the energy difference from $V = 0$ to $V = 1$ would be the same for any $V = n$ to $V = n + 1$ transition. In the realistic anharmonic model for a bond vibration this is not the case because energy levels become closer together at high V values due to a weakening of the force constant for the vibration with the higher vibrational energies. However, for lower values of V , the harmonic and anharmonic models are very similar and so we will continue our calculations using the simpler math for harmonic oscillators.)

We will use the value E_H for the ground state vibrational energy of a carbon-hydrogen bond. So $E_H = E_{V=0(H)}$ for all further discussions. We will use the value E_D for the ground state vibrational energy of a carbon-deuterium bond. So $E_D = E_{V=0(D)}$ for all further discussions.

So from Equation 10 we see the relationship between vibrational energy and frequency of the vibration, v . The frequency of vibration, v , is related to the force constant for the vibration, k (which is defined by electronic parameters and DOES NOT change with isotopic substitution), and the reduced mass of the two atoms involved in the vibration (which WILL change with isotopic substitution) by

$$v = \frac{1}{2\pi} \sqrt{\frac{k}{\mu}} \quad (11)$$

Where k is the force constant and μ is the reduced mass, which is defined by

$$\frac{1}{\mu} = \frac{1}{m_1} + \frac{1}{m_2} \quad (12)$$

Where m_1 and m_2 are the masses of the two atoms. For a C-H vibration, $m_1 = 12$ amu and $m_2 = 1$ amu. If $m_1 \gg m_2$ the Equation 12 collapses to

$$\frac{1}{\mu} \simeq \frac{1}{m_2} \quad (13)$$

And so we can say that the reduced mass is equal to the mass of the hydrogen or deuterium atom when the other atom is significantly heavier. So $\mu = m_2$ when m_1 is larger than m_2 . For hydrogen we can say that $\mu = 1$ amu and for deuterium we can say that $\mu = 2$ amu.

So lets combine Equations 10 and 11 to get

$$E_v = \frac{1}{2} h \left(\frac{1}{2\pi} \sqrt{\frac{k}{\mu}} \right) \quad (14)$$

And the vibrational energy for the unsubstituted molecules (with hydrogen, so we say that $\mu = 1$ amu) is

$$E_{v(H)} = \frac{1}{2} h \left(\frac{1}{2\pi} \sqrt{\frac{k}{1}} \right) \quad (15)$$

This can be simplified to

$$E_{v(H)} = \frac{1}{4\pi} h \sqrt{k} \quad (16)$$

And the corresponding ground state vibrational energy for a deuterium isotope substituted vibration is

$$E_{v(D)} = \frac{1}{4\pi} h \sqrt{\frac{k}{2}} \quad (17)$$

The ratio between the value of ground state vibrational energy levels in a non-deuterated molecule and a deuterated molecule is described by E_H/E_D . And so

$$\frac{E_{v(H)}}{E_{v(D)}} = \frac{\frac{1}{4\pi} h \sqrt{k}}{\frac{1}{4\pi} h \sqrt{\frac{k}{2}}} \quad (18)$$

Which simplifies to

$$\frac{E_{v(H)}}{E_{v(D)}} = \frac{\sqrt{k}}{\sqrt{\frac{k}{2}}} = \sqrt{2} = 1.41 \quad (19)$$

So we know from Equation 19 that $E_H = 1.41 E_D$.

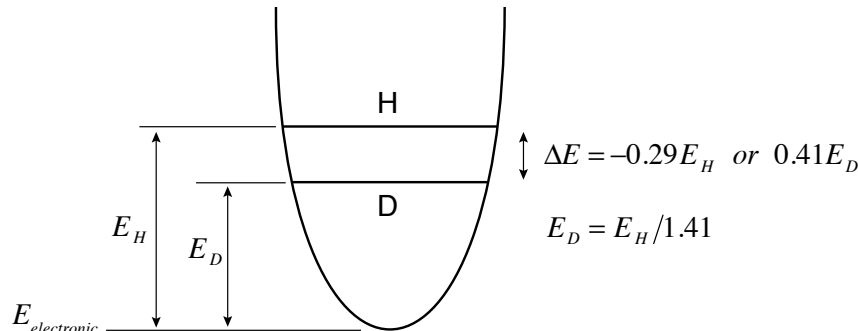


Fig. 5: Relative ground state vibrational energy levels for ^1H and ^2D substituted molecules.

Equation 19 could have been written out as

$$\frac{E_{v(H)}}{E_{v(D)}} = \frac{\sqrt{\frac{k}{m_H}}}{\sqrt{\frac{k}{m_D}}} = \sqrt{\frac{m_D}{m_H}} = \sqrt{\frac{2}{1}} = 1.41 \quad (20)$$

This is to show that the ratio between the values of the ground state vibrational energy levels with and without isotopic substitution is equal to the square root of the ratio of the isotopic masses. For a deuterium substitution for hydrogen the relative energies are described in Equation 20. For

The change in the ground state vibrational energy levels with isotopic substitution, $\Delta E_{HD} = \Delta E_D - \Delta E_H$. So we can see that, in a ^2D substitution for ^1H , ΔE_{HD} is related to the value of E_H by

$$\frac{E_H}{E_D} = 1.41$$

and so

$$\frac{E_H}{1.41} = E_D \quad (21)$$

And we know that

$$\Delta E_{HD} = \Delta E_D - \Delta E_H \quad (22)$$

Substitute Equation 21 into Equation 22 to find that

$$\begin{aligned} \Delta E_{HD} &= \frac{E_H}{1.41} - \Delta E_H \\ \Delta E_{HD} &= -\frac{0.41}{1.41} E_H = -0.29 \cdot E_H \end{aligned} \quad (23)$$

So when we substitute a hydrogen atom with a deuterium atom, the vibrational energy is lower by a value of $-0.29 \cdot E_H$. Thus the change in vibrational energy levels can be directly related to E_H for a particular molecule and E_H is directly related to the force constant for the vibration. So the higher the force constant (k), the greater the vibrational frequency (ν), the greater the ground state vibrational energy (E_H), the greater difference between the ground state vibrational energy levels between unsubstituted and isotopically substituted molecules (ΔE_{HD}).

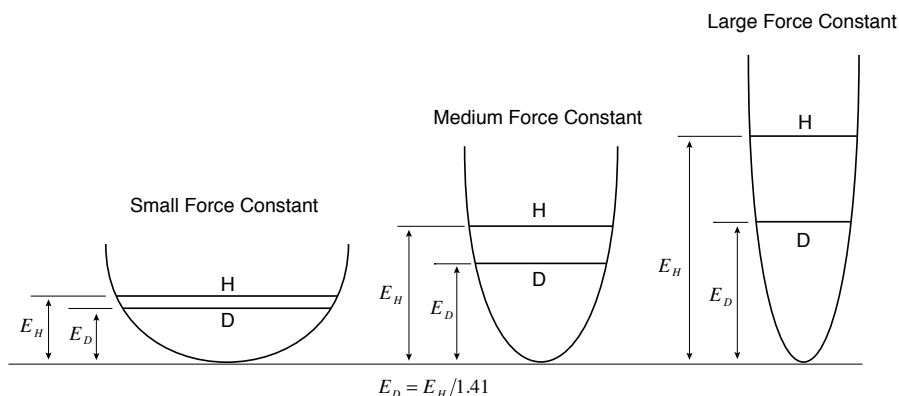


Fig. 6: Harmonic oscillators with varying force constants (k). The energy difference between vibrational energy levels in molecules with and without isotopic substitution is related to the force constant of the vibration.

So the force constant of the vibration relates directly to the vibrational energy levels for the substituted and unsubstituted molecules. And thus it relates directly to the change in vibrational energy levels on substitution by a heavier isotope.

4 The Isotope Effect Explained.

4.1 The Case of No Isotope Effect.

In an equilibrium, the equilibrium constant is defined by the energy difference between the starting material and the product. In a reaction, the rate of reaction is related to an “equilibrium” between the starting material and the transition state.

As we have seen above, the equilibrium constant is related to the change in energy between the reactant and the product molecule. In this example we will assume that the product does not have a significantly different force constant for the vibrations involving the isotopically substituted atom. In this case the

reaction coordinate for the reaction and the harmonic oscillators for the reactant and product are described in Figure 7 .

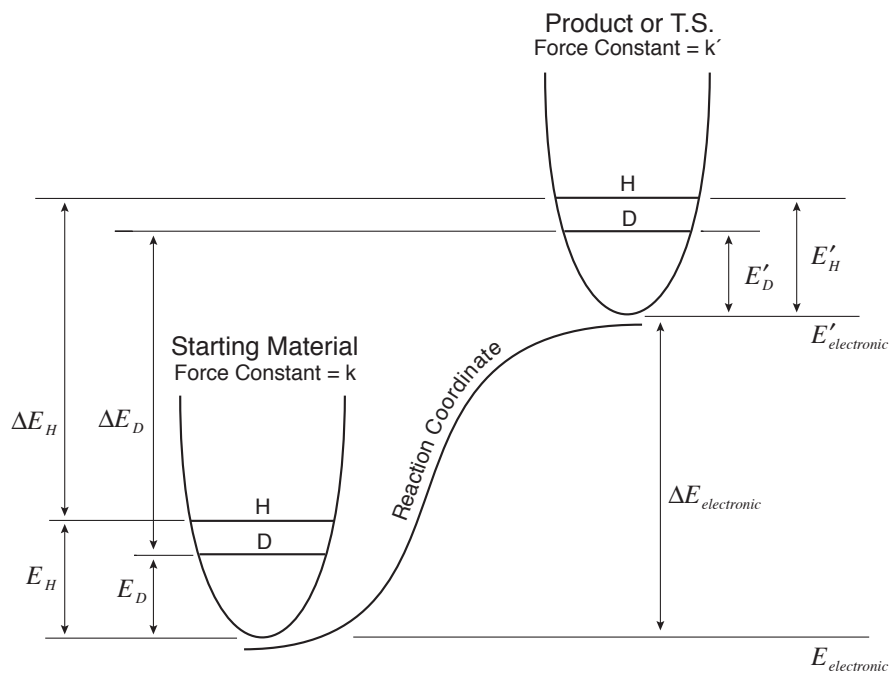


Fig. 7: Equilibrium with no change in force constant for harmonic oscillators describing vibrations involving the isotopically substituted molecules.

The energy level for the starting material is the sum of the electronic energy (E_e), the vibrational energy (E_v), the rotational energy (E_R) and the translational energy (E_T)

$$E = E_e + E_v + E_R + E_T$$

As we have seen above, the rotational and translational energy are very small contributors to overall energy and can be ignored. So we have a term for the energy of the starting material.

$$E = E_e + E_v \quad (24)$$

And the term for the energy of the product will be called E' and it is

$$E' = E'_e + E'_v \quad (25)$$

So we can see that the change in energy between the product and starting material is

$$\begin{aligned} \Delta E &= E' - E \\ \Delta E &= (E'_e + E'_v) - (E_e + E_v) \end{aligned} \quad (26)$$

Which can be rearranged to give

$$\begin{aligned}\Delta E &= (E'_e - E_e) + (E'_v - E_v) \\ \Delta E &= \Delta E_e + \Delta E_v\end{aligned}\quad (27)$$

As we know, the equilibrium constant is defined by this change in energy. Let us recall Equation 6.

$$K_{eq} = e^{\frac{-\Delta E}{R \cdot T}}$$

The isotope effect is a result of the difference in the vibrational energy levels due to isotopic substitution. As we have seen above, the isotope effect is defined by Equation 9

$$\frac{K_{eq(H)}}{K_{eq(D)}} = e^{\frac{\Delta E_D - \Delta E_H}{R \cdot T}}$$

So we need a term for ΔE_H and ΔE_D . Let us start with ΔE_H and use Equation 26.

$$\begin{aligned}\Delta E_H &= E'_H - E_H \\ \Delta E_H &= (E'_e + E'_v)_H - (E_e + E_v)_H \\ \Delta E_H &= (E'_e - E_e)_H + (E'_v - E_v)_H \\ \Delta E_H &= \Delta E_{e(H)} + \Delta E_{v(H)}\end{aligned}\quad (28)$$

and similarly

$$\Delta E_D = \Delta E_{e(D)} + \Delta E_{v(D)}\quad (29)$$

Let us define a term that describes the change in the change in energy between starting material and product (or T.S.) due to isotopic substitution.

$$\begin{aligned}\Delta\Delta E_{HD} &= \Delta E_D - \Delta E_H \\ \Delta\Delta E_{HD} &= (\Delta E_{e(D)} + \Delta E_{v(D)}) - (\Delta E_{e(H)} + \Delta E_{v(H)}) \\ \Delta\Delta E_{HD} &= (\Delta E_{e(D)} - \Delta E_{e(H)}) + (\Delta E_{v(D)} - \Delta E_{v(H)}) \\ \Delta\Delta E_{HD} &= \Delta\Delta E_{e(HD)} + \Delta\Delta E_{v(HD)}\end{aligned}\quad (30)$$

And we know by definition that the change in electronic energy is the same for both substituted and unsubstituted molecules, so we know that

$$\Delta\Delta E_e = 0$$

so

$$\Delta\Delta E_{HD} = \Delta\Delta E_{v(HD)}\quad (31)$$

So we can substitute Equation 31 into Equation 6 and say that the isotope effect is defined by

$$\frac{K_{eq(H)}}{K_{eq(D)}} = e^{\frac{-\Delta\Delta E_v}{R \cdot T}} \quad (32)$$

Now, for the example in Figure 7 on page 14 assumes that the force constants in both starting material and product are the same, so $\Delta\Delta E_v = 0$ and thus, when force constants do not change, we see that

$$\begin{aligned} \frac{K_{eq(H)}}{K_{eq(D)}} &= e^{\frac{0}{R \cdot T}} \\ \frac{K_{eq(H)}}{K_{eq(D)}} &= 1 \end{aligned}$$

So there is no isotope effect when the force constant for the vibration does not change.

4.2 Positive Isotope Effects

What if the product has a lower force constant than the starting material (e.g. the bond is getting weaker and the force constant for the stretching vibration is lower)? In this case we will have a positive isotope effect where $K_H/K_D > 1$.

We can see from the above figure that $\Delta E_{v(H)}$ is less than $\Delta E_{v(D)}$. So we know that $\Delta\Delta E_{HD}$ must be less than zero from because

$$\begin{aligned} \Delta\Delta E_{HD} &= \Delta\Delta E_{v(HD)} = \Delta E_{v(H)} - \Delta E_{v(D)} \\ \Delta\Delta E_{HD} &< 0 \text{ if } \Delta E_{v(H)} < \Delta E_{v(D)} \end{aligned}$$

If $\Delta\Delta E_{HD}$ is less than zero (negative), then the $-\Delta\Delta E_{HD}/R \cdot T$ term in Equation 32 will be positive and the exponent function of a positive number is greater than 1. Thus, when the force constant weakens in the final structure compared to the initial structure, the isotope effect will be positive and

$$\begin{aligned} \frac{K_{eq(H)}}{K_{eq(D)}} &= e^{\frac{-(negative\ value)}{R \cdot T}} = e^{\frac{positive\ value}{R \cdot T}} \\ \frac{K_{eq(H)}}{K_{eq(D)}} &> 1 \end{aligned}$$

4.3 The Maximum Value For Positive Isotope Effect.

The maximum possible value isotope effect will be when

$$\Delta\Delta E_{HD} = \Delta E_{v(HD)} \quad (33)$$

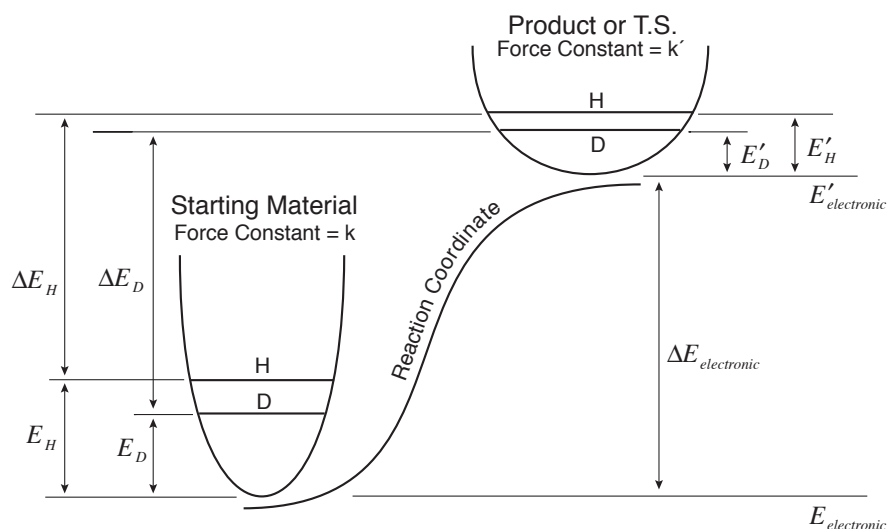


Fig. 8: Equilibrium where the force constants for harmonic oscillators describing vibrations involving the isotopically substituted molecules are different. Here, the force constant for the vibration involving the isotopically substituted atom is weaker in the product compared to the starting material.

Which is the difference in vibrational energy levels with isotopic substitution in the starting material. This can only be true when the difference in vibrational energy levels with isotopic substitution in the product is zero

$$\Delta E'_{v(HD)} = 0$$

This would be the case when the force constant in the final molecule, k' , is zero or when there is no vibrational motion in the atom being isotopically substituted in the final molecule (more on that later). See Figure on the following page.

So what is the maximum isotope effect for reactions or equilibria that involve changes in the stretching frequency of a C–H bond (e.g. C–H bond cleavage or H transfer)?

First we must calculate the E_H and E_D values. The spectroscopic data available indicates that the wavenumber for the stretching vibration of a typical C–H bond is 2900 cm^{-1} . From this value we can calculate the E_H value.

$$\begin{aligned} E_H &= \frac{1}{2} h c \bar{\nu} \\ E_H &= \frac{1}{2} (6.626 \times 10^{-34} \text{ J} \cdot \text{s}) (3.0 \times 10^{10} \text{ cm} \cdot \text{s}^{-1}) (2900 \text{ cm}^{-1}) \\ E_H &= 2.88 \times 10^{-20} \text{ J} \end{aligned}$$

This is the value for a single molecule. For a mole of molecules we must multiply

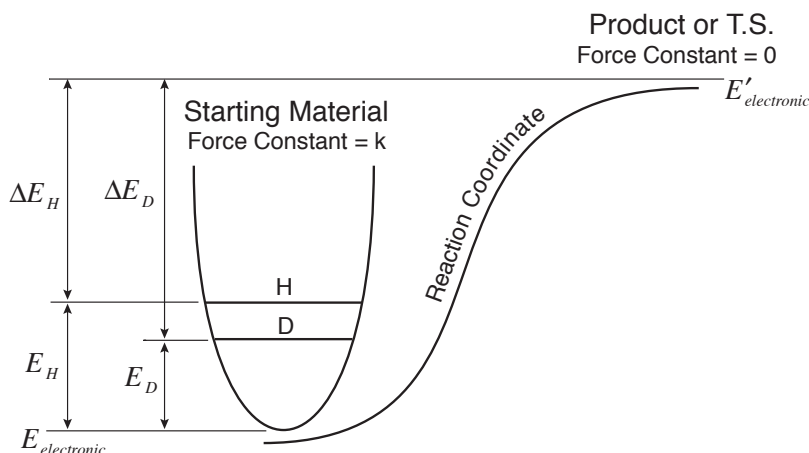


Fig. 9: Maximum isotope effect when $\Delta\Delta E_{HD} = \Delta E_{v(HD)} = \Delta E_H - \Delta E_D$.

by Avagadro's number.

$$\begin{aligned} E_H &= N \cdot (2.88 \times 10^{-20} \text{ J}) \\ E_H &= (6.023 \times 10^{23} \text{ mole}^{-1}) (2.88 \times 10^{-20} \text{ J}) \\ E_H &= 17,360 \text{ J} \cdot \text{mole}^{-1} \end{aligned}$$

And we know from Equation 19 that

$$\frac{E_H}{E_D} = 1.41$$

So $E_D = E_H/1.41$ and $\Delta E_{HD} = -0.29 \cdot E_H$ (from Equation 23)

Using the above relationship it can be calculated that the wavenumber for the vibration of a C—D bond should be 2057 cm^{-1} . This figure agrees well with the experimentally observed value of 2085 cm^{-1} .

So $\Delta E_{HD} = -0.29 \cdot E_H$ and for the maximum isotope effect we have stated that $\Delta\Delta E_{v(HD)} = \Delta E_{HD}$, so Equation 32 becomes

$$\frac{K_{eq(H)}}{K_{eq(D)}} = e^{\frac{0.29 \cdot E_H}{R \cdot T}} \quad (34)$$

Where R is the gas constant ($R = 8.314 \text{ J} \cdot \text{mole}^{-1} \cdot \text{K}^{-1}$) and T is the temperature in Kelvin (lets say $T = 298 \text{ K}$).

So the maximum positive primary isotope effect we would expect to see would be about 7.6.

4.4 Inverse Isotope Effects

When the force constant for the vibrations involving the isotopic substitution is larger in the product or transition state than in the starting material we will have a situation where $\Delta E'_D$ is greater than ΔE_H . See Figure 10.

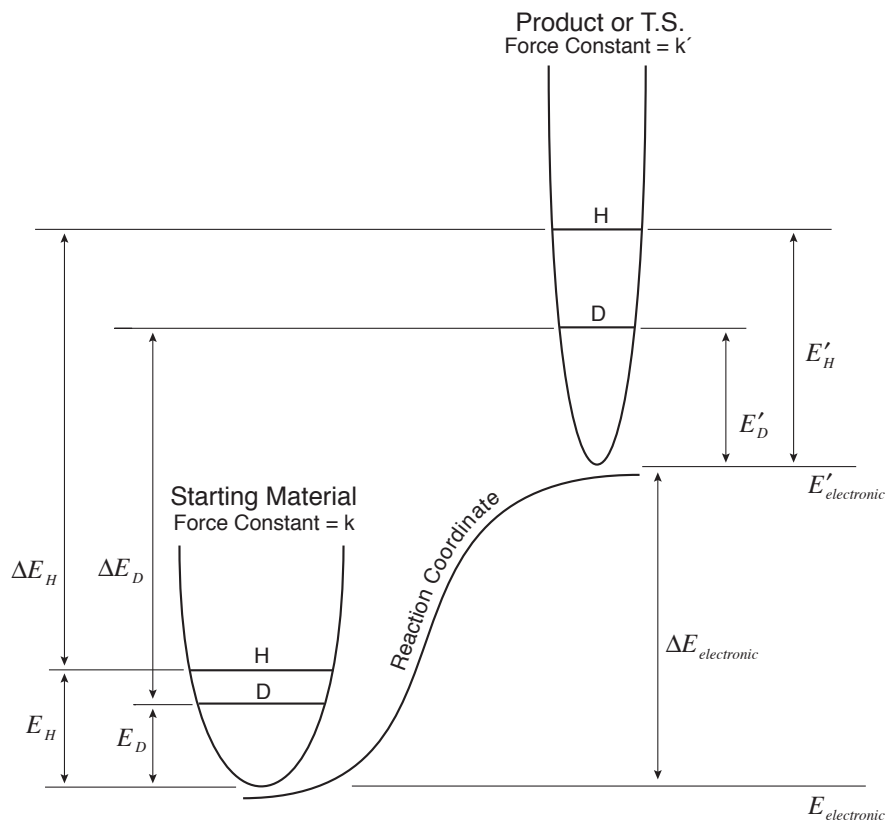


Fig. 10: Inverse isotope effect when the force constant in the product, k' , is greater than the force constant in the starting material, k .

We can see from the above figure that $\Delta E_{v(H)}$ is greater than $\Delta E_{v(D)}$. So we know that $\Delta\Delta E_{HD}$ must be greater than zero because

$$\Delta\Delta E_{HD} = \Delta\Delta E_{v(HD)} = \Delta E_{v(H)} - \Delta E_{v(D)}$$

$$\Delta\Delta E_{HD} > 0 \text{ if } \Delta E_{v(H)} > \Delta E_{v(D)}$$

If $\Delta\Delta E_{HD}$ is less than zero (negative), then the $-\Delta\Delta E_{HD}/RT$ term in Equation 32 will be positive and the exponent function of a positive number is greater

than 1. Thus, when the force constant increases in the final structure compared to the initial structure, the isotope effect will be inverse and

$$\frac{K_{eq(H)}}{K_{eq(D)}} = e^{\frac{-positive\ value}{R \cdot T}} = e^{negative\ value}$$
$$\frac{K_{eq(H)}}{K_{eq(D)}} < 1$$

5 Kinetic Isotope Effect in Chemistry

We can use the magnitudes of the values for primary and secondary kinetic isotope effects to interpret the mechanism of chemical reactions

5.1 Elimination Reactions.

Does the mechanism of a particular elimination reaction follow an E2, E1 or E1cb mechanism?

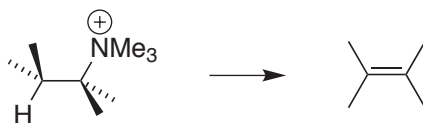
The E2 Mechanism. A concerted elimination where the leaving group and the hydrogen depart nearly simultaneously as shown in Figure 11 on the following page. The hydrogen is removed by the concerted attack of a base.

The E1 Mechanism. Like the SN1 mechanism, the RDS involves the departure of leaving group to give a carbocation as shown in Figure 12 on page 22. The proton is removed by the base in the subsequent step.

The E1cb Mechanism. The hydrogen departs first with the attack of a base to produce a carbanion as shown in Figure 13 on page 23. This is the R.D.S. Then the leaving group departs.

So we can use our knowledge of the transition state structures for the RDS of each mechanism and any kinetic isotope effects observed for the reaction.

For the following reaction which occurs in the presence of base...



Where L is the hydrogen or deuterium atom. We observe the following kinetic isotope effects...

$$\frac{k_H}{k_D} = 4.6 \text{ and } \frac{k_{^{14}\text{N}}}{k_{^{15}\text{N}}} = 1.009$$

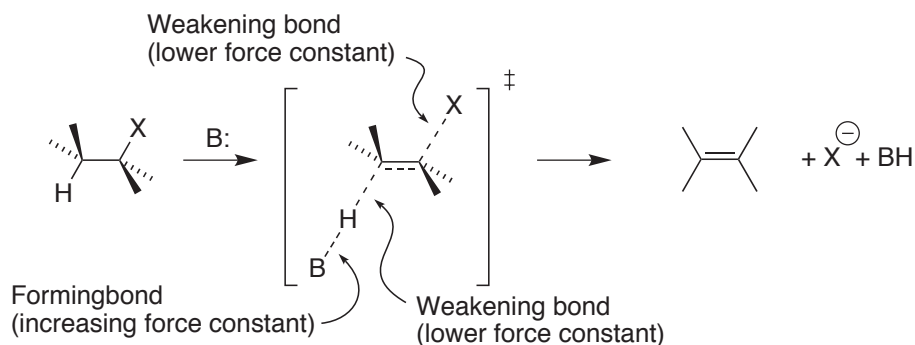


Fig. 11: The E2 elimination reaction. Both the base and the substrate are involved in the R.D.S.

The values for kinetic isotope effects for deuterium substitution of hydrogen and ^{15}N substitution of ^{14}N are both large enough to be considered primary kinetic isotope effects. The maximum values for primary isotope effects are ~ 7 for $^1H/^2D$ and for 1.03 for $^{14}N/^{15}N$.

Since we have significant PKIE's for both the proton and the nitrogen of the leaving group we can conclude that both bonds are breaking in the T.S. of the R.D.S. Thus we can conclude that the most likely mechanism is the E2 mechanism.

Challenge: Can you suggest an experiment using Hammett σ values and rates of reaction that would also probe this mechanism? What results (ρ values) would you expect for each mechanism?

Tab. 2: Maximum Theoretical Values for Primary Kinetic Isotope Effects.

Isotopic Substitution	Maximum Value of PKIE
$^1H/^2D$	6.4
$^{12}C/^{13}C$	1.04
$^{12}C/^{14}C$	1.07
$^{14}N/^{15}N$	1.03
$^{16}O/^{18}O$	1.02
$^{32}S/^{34}S$	1.01
$^{35}Cl/^{37}Cl$	1.01

Let us consider that the kinetic isotope effect experiments gave the following values (these are fictional values for consideration)...

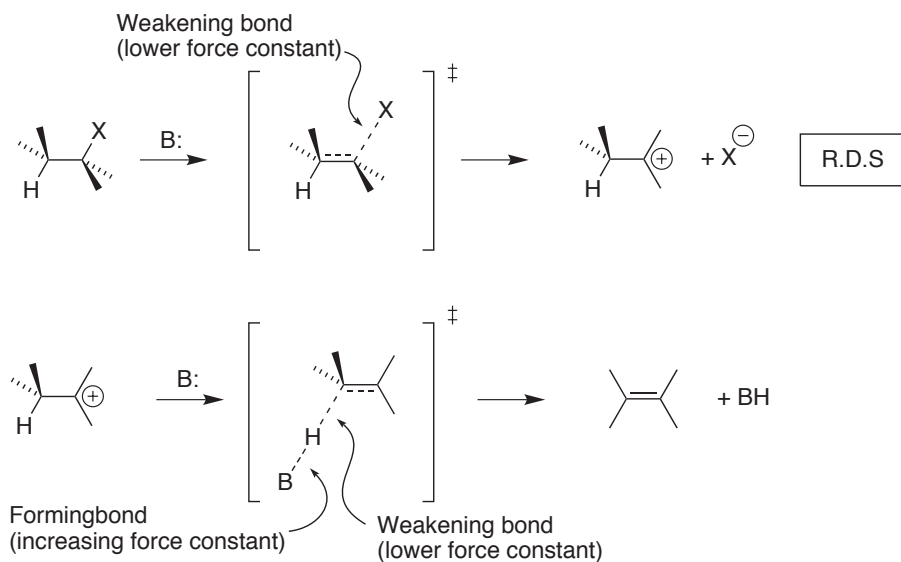


Fig. 12: The E1 mechanism. The leaving group departs in the R.D.S. Only after formation of the cation is the substrate acidic enough for the base to remove the proton and complete the reaction.

$$\frac{k_H}{k_D} = 4.6 \text{ and } \frac{k_{^{14}\text{N}}}{k_{^{15}\text{N}}} = 1.0000$$

We see that C—H bond breakage is involved in the T.S. of the R.D.S. but that C—N bond cleavage is not important. Thus we can say that the E1cb mechanism is the most likely mechanism to explain these results.

What if the values for the kinetic isotope effects were...

$$\frac{k_H}{k_D} = 1.2 \text{ and } \frac{k_{^{14}\text{N}}}{k_{^{15}\text{N}}} = 1.013$$

The value for the $^1\text{H}/^2\text{D}$ substitution is too small to be a primary kinetic isotope effect but the value for $^{14}\text{N}/^{15}\text{N}$ substitution is large enough to be a PKIE. So we can conclude that breakage of the C—N bond is important in the T.S. of the R.D.S. and that C—H breakage is not. The value for $^1\text{H}/^2\text{D}$ substitution is due to the influence of the change in mass of the molecule at one end of the C—N vibration. It is a secondary isotope effect.

From the above reasoning we can conclude that the E1 mechanism is the most likely mechanism to explain these results.

For the reaction shown in Figure 14 on the next page, we see that

$$\frac{k_H}{k_D} = 1.1$$

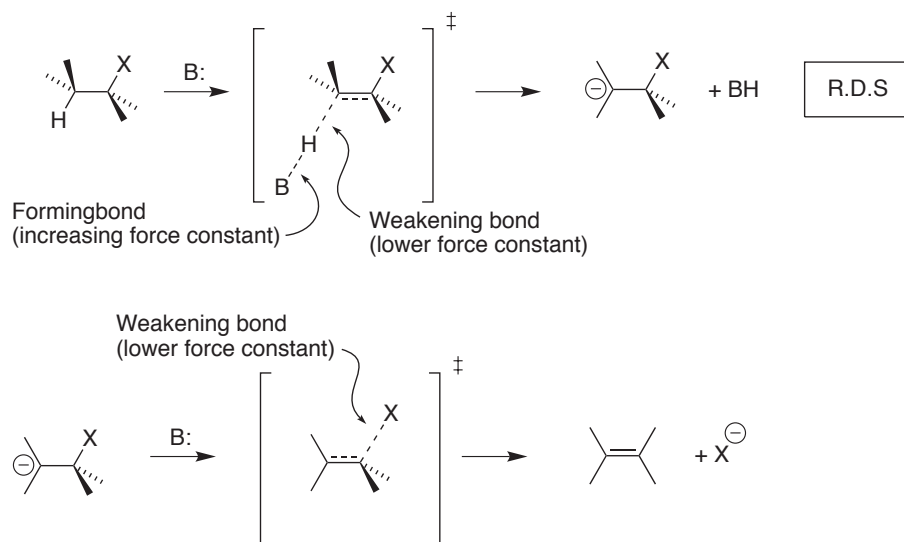


Fig. 13: The $E1_{cb}$ reaction. The leaving group is very poor and so the elimination can only occur if a strong base creates a carbanion which can then eliminate the leaving group.

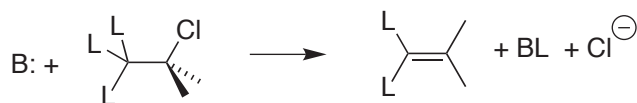


Fig. 14: An example reaction. L can be hydrogen or deuterium

This value is too small to be due to PKIE's and so we can reject mechanism that involve C—H breakage in the T.S. of the R.D.S. ($E2$ and $E1_{cb}$) thus leading us to conclude that the $E1$ mechanism is the most likely.

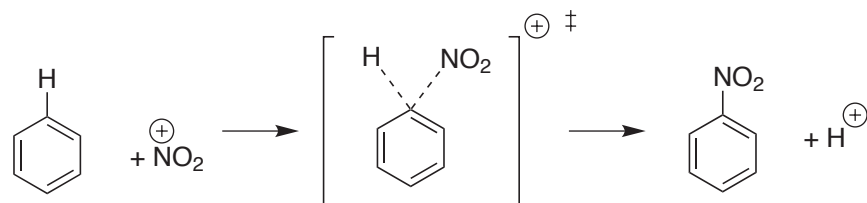
5.2 Nitration of Aromatic Ring Systems

The nitration of a benzene ring can be imagined to proceed via one of two possible mechanisms shown in Figure 15 on the following page. But which is correct?

When we use deuterated benzene in this reaction we see that...

$$\frac{k_H}{k_D} = 1.000$$

This is strong evidence that the C—H vibration does not change bond strength



(a) Concerted Addition-Elimination

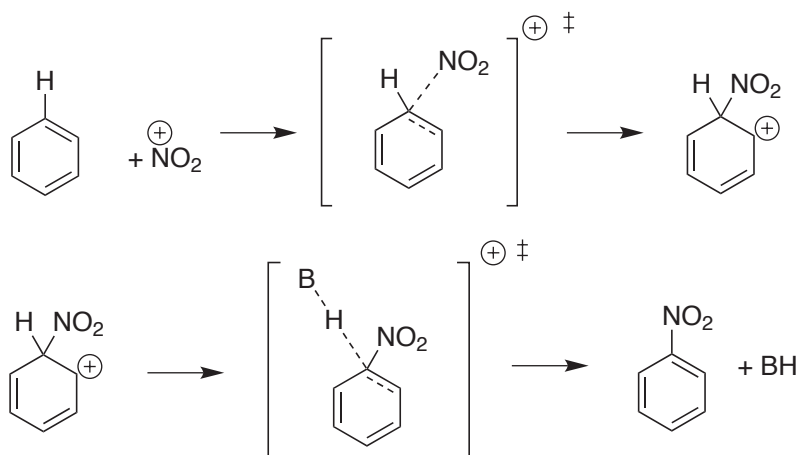
(b) Electrophilic Attack and Subsequent Departure of Hydrogen (S_N2Ar)

Fig. 15: Two possible mechanisms for nitration of an aromatic ring.

or length during the R.D.S. so we can conclude that the reaction proceeds via the S_N2Ar mechanism described in Figure 15.

5.3 Secondary Effects and Mechanism

The secondary kinetic isotope effect (SKIE) is due to changes in the vibrations important in the T.S. of the R.D.S. because of mass changes due to isotopic substitution but these vibrations do not directly involve the substituted atom.

For example, in the S_N1 reaction for the nucleophilic substitution of alkyl bromides the R.D.S. is the dissociation of the bromine to give the carbocation intermediate (see Figure 16 on the next page). Substitutions of hydrogens for deuterium cannot result in PKIE's because only the C—Br bond is changing during the R.D.S. However, we will see SKIE's due to mass changes in the molecule. These effects are cumulative per substitution and are often considered to be “percentage” effects (which simplifies the math for the cumulative effects)

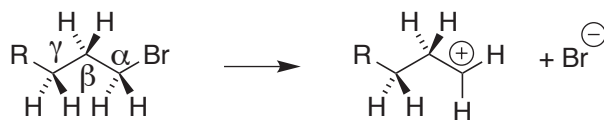


Fig. 16: Formation of a carbocation

Table 3 reports typical SKIE's for single substitutions at the α , β and γ carbon positions of the reactant (positions are relative to the cationic carbon in Figure 16)

Tab. 3: Secondary Kinetic Isotope Effects for $^1\text{H}/^2\text{D}$ Substitutions

Position	$k_{\text{H}}/k_{\text{D}}$	% per D
α	1.1 to 1.2	10 to 20
β	1.15 to 1.25	15 to 25
γ	0.92 to 1.02	-8 to 2

The effect at the γ position is small as might be expected. However the effect of deuterium substitutions at the β position is larger than at the α position. This is unexpected, as we would think that the isotope effect would diminish with distance.

5.3.1 Hyperconjugation and Secondary isotope Effects

Hyperconjugation is one way of conceptualizing the mechanism for the inductive electron donating effect of methyl groups. We say that the C—H bonds in the methyl group are donating their bonding electrons into the carbon center or adjacent bond via resonance as shown in Figure 17.

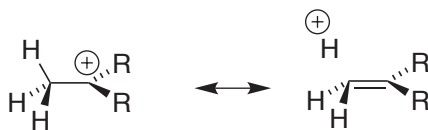


Fig. 17: Hyperconjugation in a carbocation

The effect is especially obvious in $S_{\text{N}}1$ and $E1$ mechanisms where carbocations are stabilized by the hyperconjugation as seen in the resonance system above.

When a C—H bond is parallel to the empty P_z orbital of the SP_2 hybridized carbocation the P_z orbital on the adjacent carbon, which is a contributor to the

SP_3 orbital on the carbon that is involved in the C—H bond, can donate a fraction of its electron density into the empty orbital to give a slight orbital overlap which helps to stabilize the carbocation (see Figure 18). Since hyperconjugation requires that a C—H bond be broken in the contributing resonance structure we will see a slight PKIE mixed in with the SKIE for β -isotopic substitutions.

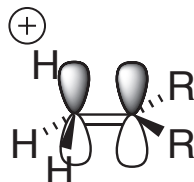


Fig. 18: Orbital overlap model for hyperconjugation

A C—H bond must be rotated into a parallel position for this to happen. This is easy to achieve in an aliphatic molecule but the effect has obvious consequences when we restrict rotation, such as in ring systems. As shown in Figure 19 on the following page, a C—D bond at an angle of near 0° with respect to the P_Z orbital of the carbocation (near perpendicular to the SP_2 plane) will give the greatest isotope effect at the β -carbon. An angle of near 90° with respect to the P_Z orbital of the carbocation (near parallel to the SP_2 plane) will give the least isotope effect at the β -carbon.

5.3.2 Magnitude of Secondary Isotope Effect.

The change in the vibrational ground state energy levels for deuterated and non-deuterated molecules involved in an S_N1 or $E1$ reaction mechanism is a good example of the SKIE. The molecule must change from SP_3 to SP_2 with corresponding changes in the force constant for the C—H vibrations involved (wagging vibrations are the most important as the H—C—H bond angles must change during the reaction so the wagging vibrations best represent the path of the reaction coordinate for the H—C—H angle).

Using spectroscopic data we can determine the vibrational energies for representative SP_3 and SP_2 C-H wagging vibrations.

In the diagrams in the table, the in-plane (of the paper) bending vibrations best reflect the changes to the C—H bonds in the transition from SP_3 to SP_2 . In the transition from SP_3 to SP_2 , we see that the force constant is getting smaller so we expect a normal isotope effect.

The maximum isotope effect is defined by the change in the change in vibrational energies from reactant to product with isotopic substitution, $\Delta\Delta E_{v(HD)}$. We know that

$$\Delta\Delta E_{v(HD)} = \Delta E_{v(H)} - \Delta E_{v(D)}$$

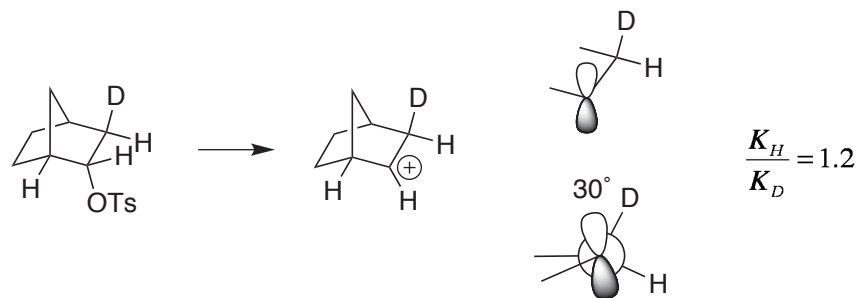
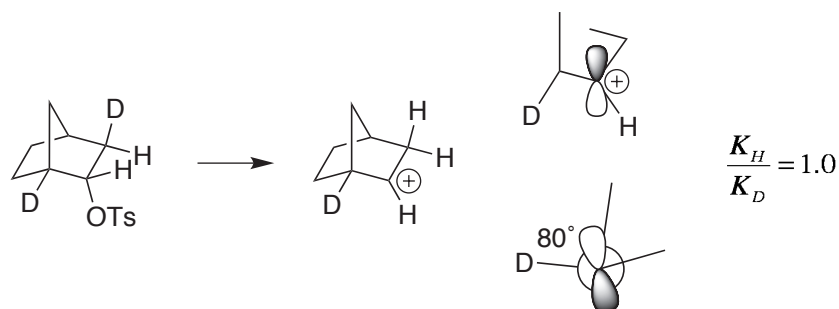
(a) The C—D bond parallel to the empty P_z orbital.(b) The C—D bond is perpendicular to the empty P_z orbital.

Fig. 19: A Norbornane system demonstrates the effect of the dihedral angle between the C—D bond and the empty P_z orbital in the cation on the magnitude of the secondary isotope effect.

and

$$\begin{aligned}\Delta E_{v(H)} &= \frac{1}{2}hc\bar{\nu} - \frac{1}{2}hc\bar{\nu}' \\ \Delta E_{v(H)} &= \frac{1}{2}hc(\bar{\nu} - \bar{\nu}') \\ \Delta E_{v(H)} &= \frac{1}{2}hc \cdot \Delta\bar{\nu}\end{aligned}$$

where $\bar{\nu}$ is the wavenumber for the C—H bending in starting material and $\bar{\nu}'$ is the wavenumber for the C—H bending in product.

Similarly, we see that $\Delta E_{v(D)} = \frac{1}{2}hc \cdot \Delta\bar{\nu}_{(D)}$ using the wavenumbers for the

Tab. 4: Stretching Frequencies of C—H Bonds

		
$\bar{\nu}$ of C—H stretch	2900 cm^{-1}	2800 cm^{-1}
$\bar{\nu}$ of C—H stretch	2085 cm^{-1}	2020 cm^{-1}
$\bar{\nu}$ of C—H bending	1340 & 1340 cm^{-1}	1350 & 800 cm^{-1}
$\bar{\nu}$ of C—D bending	957 cm^{-1}	571 cm^{-1}

1340 cm^{-1} is vibration for in-plane (of paper) and out-of-plane (of the paper) bending vibrations in SP_3 C—H bonds. 1350 cm^{-1} is vibration for out-of-plane (of the paper) vibrations for C—H bonds in SP_2 carbonds and 800 cm^{-1} is vibration for in-plane (of the paper) vibrations for C—H bonds in SP_2 carbons. The C—D value are both for the in-plane bending vibration.

deuterium substituted molecules. So we see that

$$\begin{aligned}
 \Delta\Delta E_{v(HD)} &= \frac{1}{2}hc \cdot \Delta\bar{\nu}_{(H)} - \frac{1}{2}hc \cdot \Delta\bar{\nu}_{(D)} \\
 \Delta\Delta E_{v(HD)} &= \frac{1}{2}hc \cdot (\Delta\bar{\nu}_{(H)} - \Delta\bar{\nu}_{(D)}) \\
 \Delta\Delta E_{v(HD)} &= \frac{1}{2}hc \cdot ((1340 - 957) - (800 - 571)) \\
 \Delta\Delta E_{v(HD)} &= \frac{1}{2}hc \cdot (154 \text{ cm}^{-1}) \\
 \Delta\Delta E_{v(HD)} &= 1.53 \times 10^{-21} \text{ J}
 \end{aligned}$$

Of course this value is for a single molecule. We must multiply by Avagadro's number to get the change in the change in energy per mole.

$$\begin{aligned}
 \Delta\Delta E_{v(HD)} &= (1.53 \times 10^{-21} \text{ J}) \cdot (6.023 \times 10^{23} \text{ mole}^{-1}) \\
 \Delta\Delta E_{v(HD)} &= 921 \text{ J} \cdot \text{mole}^{-1}
 \end{aligned}$$

And we have calculated this value from E_D to E_H . For the isotope equation we are looking at the transition from the direction of E_H to E_D so we must take the negative to get $\Delta\Delta E_{v(HD)} = 921 \text{ J} \cdot \text{mole}^{-1}$.

And we have see that the isotope effect can be calculated from $\Delta\Delta E_{v(HD)}$ by using

$$\begin{aligned}
 \frac{k_H}{k_D} &= e^{-\frac{\Delta\Delta E_{v(HD)}}{R \cdot T}} \\
 \frac{k_H}{k_D} &= e^{-\frac{-921}{8.314 \cdot 298}} \\
 \frac{k_H}{k_D} &= 1.45
 \end{aligned}$$

So from this simplified approach we see that the maximum SKIE we might expect for carbocation formation with isotopic substitution at the α carbon is 1.45. In reality the effect can be slightly larger than this value (because the substitution will affect more vibrations than the one we considered above) but in practice the observed values for SKIE seldom exceeds 1.2.

6 Kinetic Isotope Effect in Chemistry 2

6.1 Transition State Geometry.

Consider the *E1cb* reaction mechanism in Figure 20.

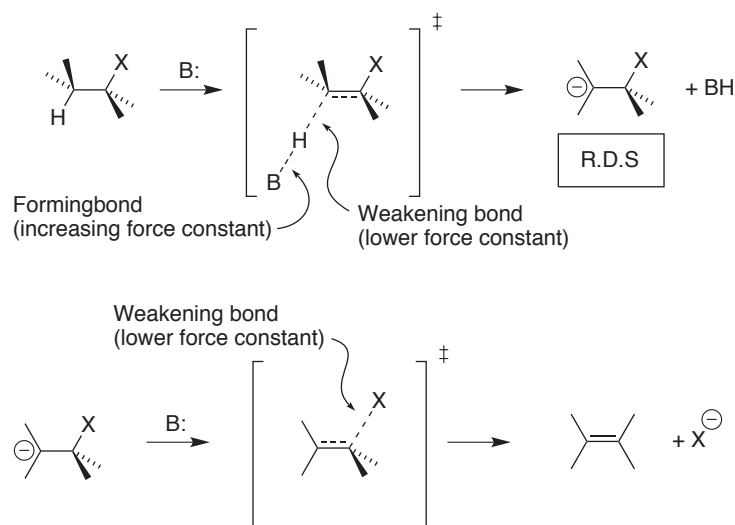


Fig. 20: The two steps of the *E1cb* reaction mechanism. The leaving group is poor so a strong base is required to remove the proton as the first and rate-determining step. After the anion is formed the α -elimination of the leaving group is rapid.

6.1.1 Symmetrical Transition States

The reaction coordinate might resemble the schematic diagram in Figure 22 on page 31. Note that the free energy of the initial condition (base and reactant) are comparable with the free energy of the products of the R.D.S. (protonated base and deprotonated reactant). According to the Hammond postulate, which states that the transition state will most resemble the higher energy structure in the reaction coordinate, we can surmise that the transition state falls fairly evenly between the two states in the reaction coordinate. Thus the proton is

about half-way along its journey from the carbon atom of the reactant to the basic atom of the base. The transition state has a symmetrical arrangement for the Base—H—C structure (see Figure 21).

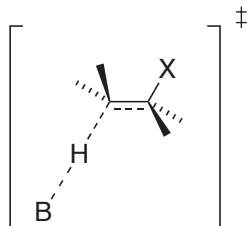


Fig. 21: The transition state for the R.D.S. of the E1cb elimination mechanism.

When we have a symmetrical transition state the stretching vibrations in the transition state will no longer be affected by changes in the mass of the proton (e.g. substituting it with a deuterium). This is because in a symmetrical T.S. the proton is not moving in the stretching vibration that is involved in defining the vibrational energy for the structure, the symmetrical stretching vibration (see Figure 23 on page 32)

When the proton does not move, there is no effect on vibrational energy levels in the structure due to isotopic substitution of that proton. Thus the change in the energy difference between the vibrational energy levels due to isotopic substitution in the starting condition will not be lessened by corresponding energy differences in the T.S. There are no energy differences due to isotopic substitution in the T.S. so the isotope effect will be at a maximum.

6.1.2 'Early' Transition States.

Let us consider the situation in Figure 25 on page 33 where the starting materials are collectively high in energy (e.g. when B is a very strong base and is high in energy without a proton or the reactant is a strong acid and is high in energy with a proton)

In this case the transition state closely resembles the starting material. So there has not been a great change in the nature of the C—H bond (it is only just beginning to change at the highest energy point of the coordinate). Thus the vibrational energy levels for hydrogen or deuterium substitution will be fairly similar in the initial condition and in the T.S. This will result in a low value for the kinetic primary isotope effect.

In the extreme case, where the T.S. so closely resembles the starting conditions that they are virtually identical, there will be no isotope effect.

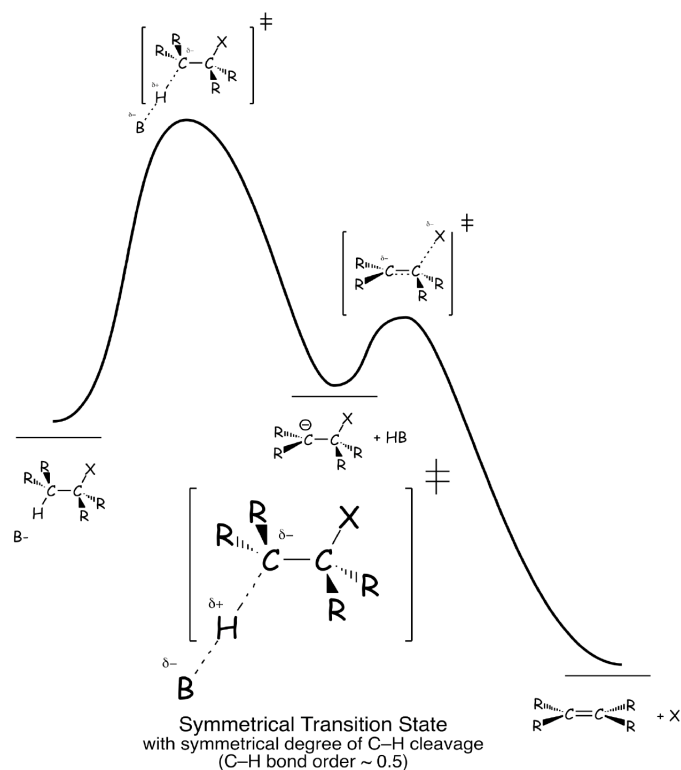


Fig. 22: An example reaction coordinate diagram for the E1cb elimination mechanism with a symmetrical transition state in its R.D.S.

6.1.3 'Late' Transition States

In the case of a late transition state shown in Figure 27 on page 35, we will also have a low isotope effect.

The T.S. involves a strong C—H bond which, if the basic atom is carbon, will have the same force constant as in the reactant. Thus the force constant for the C—H vibration in the T.S. is the nearly the same as in the C—H vibration of the reactants. the effect on vibrational energy levels of the T.S. for isotopic substitution will be the same as in the reactants and we will observe a small isotope effect.

In the limiting case where the T.S. structure is virtually identical to the products we will have no isotope effect. If the basic atom is not carbon we will have a different force constant but it will not be zero and so we will expect to see a small isotope effect in the limiting case above.

So at either extreme for the location of the transition state we will see small isotope effects and in the symmetrical T.S. we will see isotope effects near the

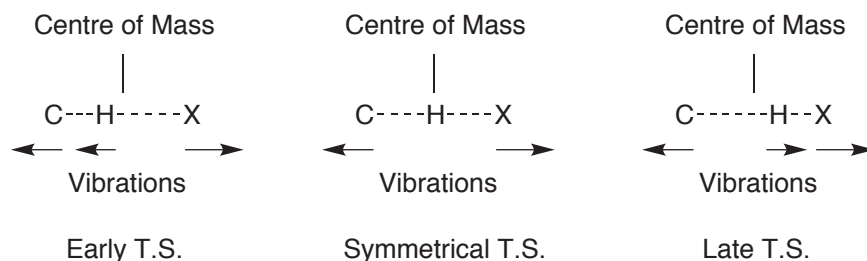


Fig. 23: Stretching vibrations in early, symmetrical and late transition states for proton transfer.

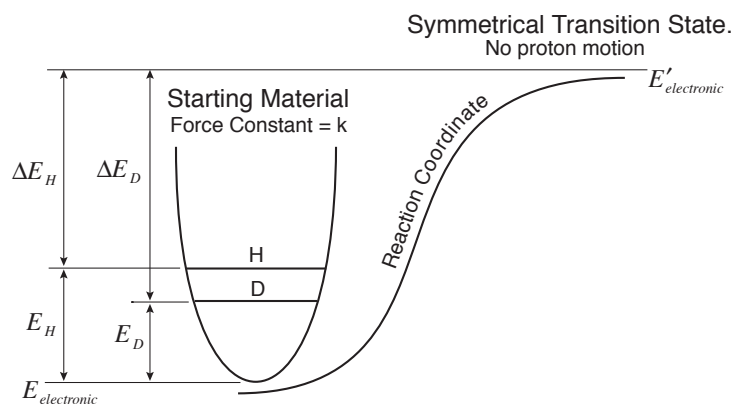


Fig. 24: Maximum isotope effect in symmetrical transition state for proton transfer (compare to Figure 9 on page 18).

maximum possible. Depending on the location of the T.S. in the reaction coordinate we will see different magnitudes for the isotope effect as demonstrated in Figure 28 on page 36.

6.1.4 Relative Basicity and isotope Effect

The location of the T.S. can also be interpreted as a consequence of the relative basicities of the base and the conjugate base of the reactant (carbanion). If they are comparable and the difference in the pK_a s of the two bases is small we will have a symmetrical T.S. as the energies of the reactants and products will be comparable (see Figure 22 on the preceding page). If one base is much stronger than the other there will be a high energy situation when that base is deprotonated and the T.S. will resemble the initial or final conditions (see Figures 25 and 27) depending on which base is the stronger. Figure 29 on page 37 demonstrates this effect.

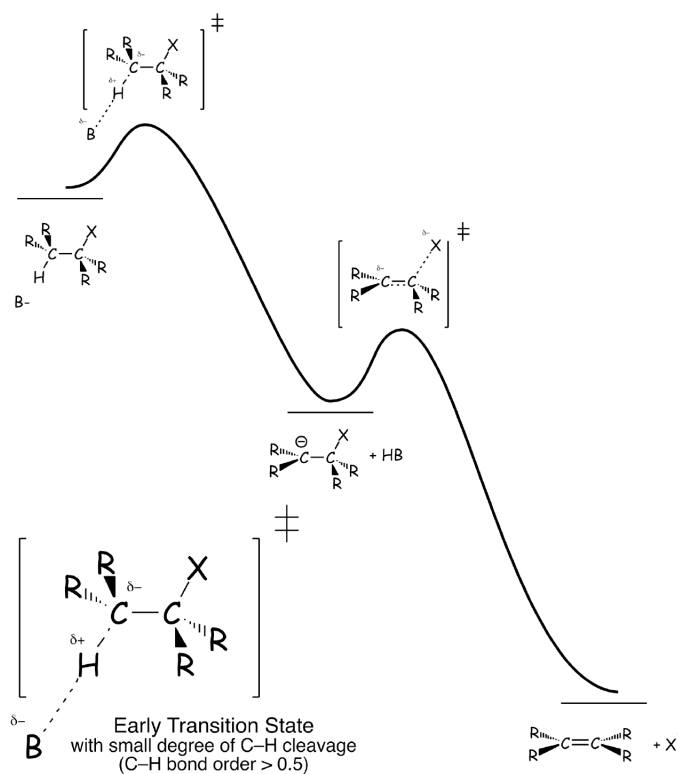


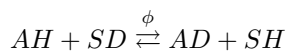
Fig. 25: Reaction coordinate for E1cb reaction with early T.S. in the R.D.S

6.2 Solvent Isotope Effects

When we isotopically substitute atoms in a solvent we will see isotope effects in reactions where solvent participates in the R.D.S. the most common example is the effect of D_2O in reactions involving acid/base equilibria or proton exchange with water in the R.D.S.

The water solvent isotope effect is expressed using the concept of a 'fractionation factor', ϕ , which is the equilibrium constant for the exchange of a given proton for a deuterium in a molecule. We will write the reactant as A—H, the deuterated solvent as S—D, the exchanged deuterated product as A—D and the exchanged protonated solvent as S—H.

For the equilibrium the following equilibrium...



We know that

$$\phi = \frac{[AD][SH]}{[AH][SD]} \quad (35)$$

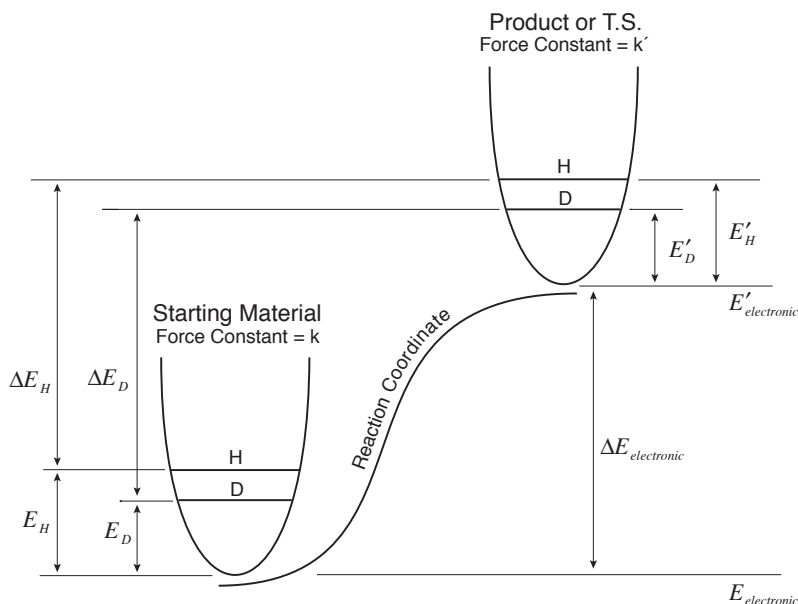


Fig. 26: Limiting case where starting material and transition state are very similar and there is no isotope effect because the C—H bond strength (force constant) is the same for both reactant and transition state.

The equilibrium isotope effect between 100% D₂O and 100% H₂O is given by

$$\frac{K_{D_2O}}{K_{H_2O}} = \frac{\phi_p}{\phi_r} \quad (36)$$

Where ϕ_r is the fractionation factor for the reactant and ϕ_p is the fractionation factor for the product.

The kinetic isotope effect between 100% D₂O and 100% H₂O is given by

$$\frac{K_{D_2O}}{K_{H_2O}} = \frac{\phi_{\mp}}{\phi_r} \quad (37)$$

Where ϕ_r is the fractionation factor for the reactant and $\phi_{\mp}$ is the fractionation factor for the product.

A reactant with more than one proton will have more than one fractionation factor. We must take the product of the fractionation factors for each proton to get the true isotope effect so Equation 36 becomes...

$$\frac{K_{D_2O}}{K_{H_2O}} = \frac{\prod_i \phi_p^i}{\prod_i \phi_r^i} \quad (38)$$

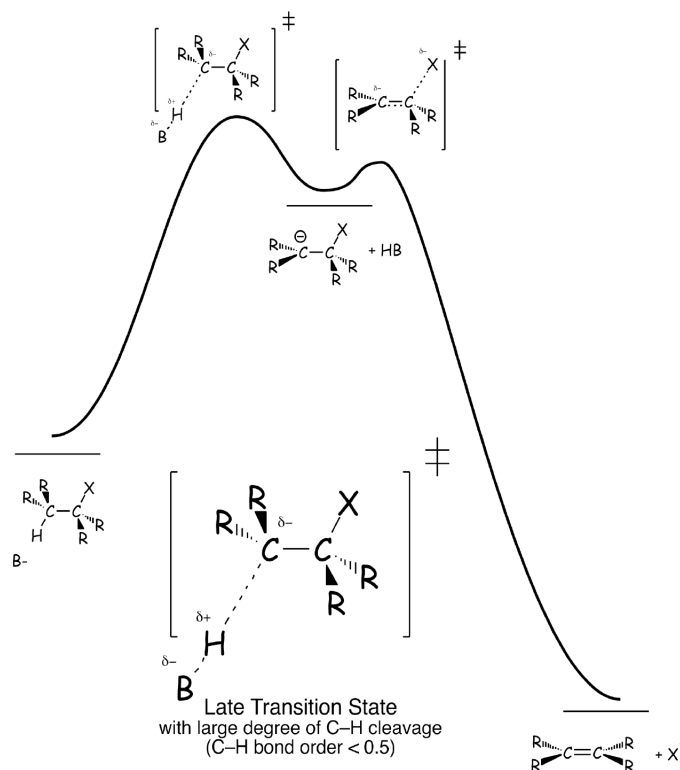
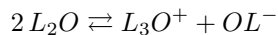


Fig. 27: Reaction coordinate for $E1cb$ reaction with late T.S. in the R.D.S

where we have taken the product for all fractionation factors in reactant and product (or transition state)

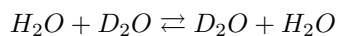
6.2.1 Solvent Effects for Acid Dissociation of Water

We consider the equilibrium...



where L can be either H or D. We must use the fractionation factors for each of the terms. L_2O will have two identical fractionation factors, ϕ_{L_2O} , L_3O^+ will have three identical fractionation factors, ϕ_{L_3O} , and OL^- will have only one fractionation factor, ϕ_{OL} .

ϕ_{L_2O} is by definition 1.0 because for the equilibrium



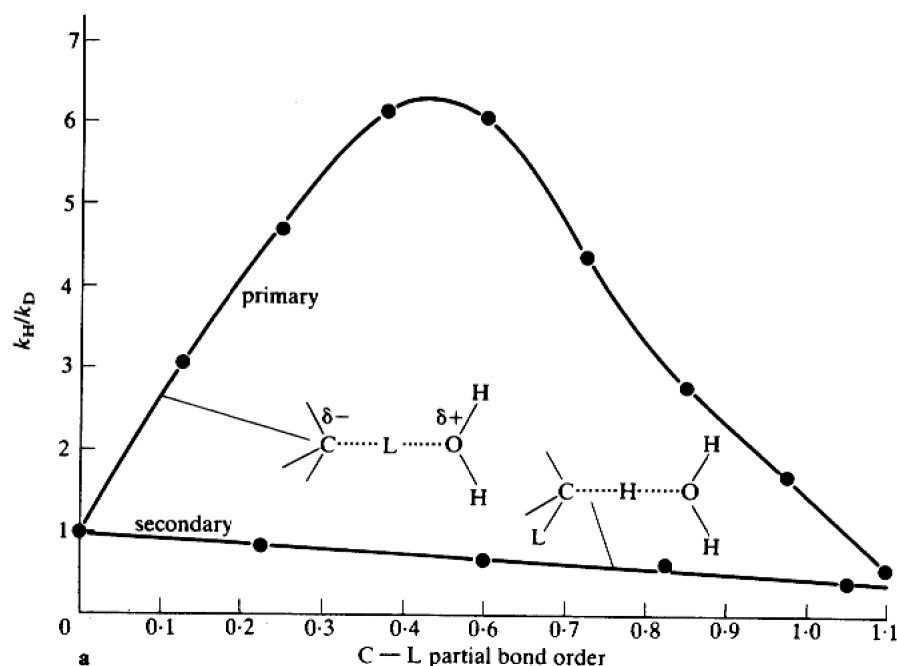
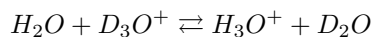


Fig. 28: (Fig. 7.2. from Isaacs) The magnitude of primary isotope effect compared to location of transition state as described by C—H bond order (1.0 means C—H bond is unaffected, 0.5 means we are halfway to breaking and in a symmetrical T.S. and 0 means the C—H bond is completely broken in the T.S.) Note: The bond order if 1.1 is due to the fact that the bond order is measured by bond length and the C—D bond is slightly shorter than the C—H bond and therefore of a apparently higher order.

we have

$$\phi_{L_2O} = \frac{[D_2O][H_2O]}{[H_2O][D_2O]} = 1 \quad (39)$$

ϕ_{L_3O} has been experimentally determined to be 0.69



$$\phi_{L_2O} = \frac{[D_3O^+][H_2O]}{[H_2O][D_3O^+]} = 0.69 \quad (40)$$

The equilibrium above favours the reactants because, in equilibria, the heavier isotope will favour being in the structure with the larger force constant. L_2O has a larger force constant than L_3O^+ because of the greater electron density (the third proton or deuteron will diminish electron density at the oxygen). So the deuteron will prefer to be in the L_2O molecule rather than the L_3O^+ molecule.

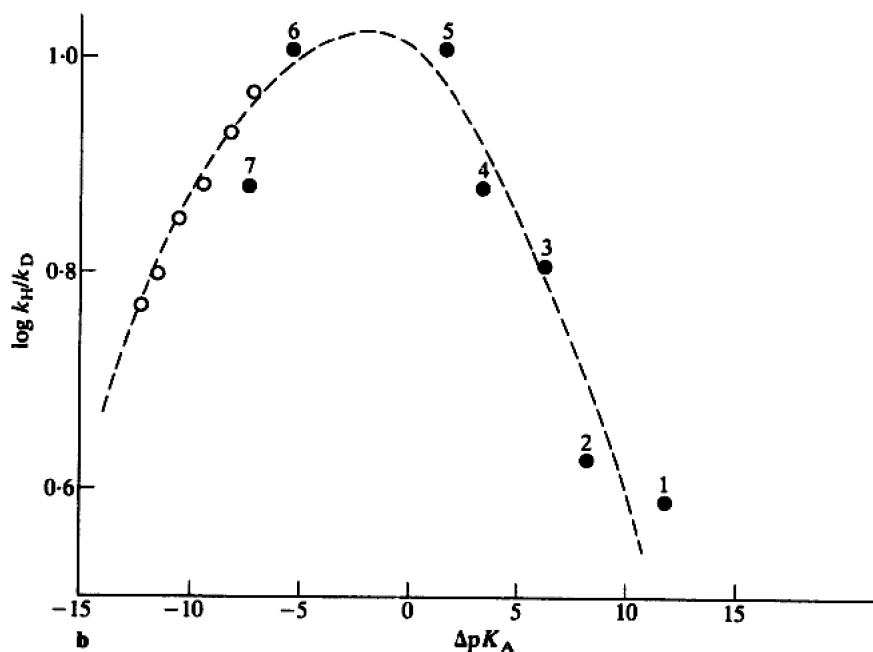


Fig. 29: Isotope effect for proton transfer vs. ΔpK_a values for base and conjugate base of reactant [$\Delta pK_a = pK_a(\text{reactant}) - pK_a(\text{base})$]

The kinetic isotope effect for acid dissociation of water is experimentally determined to be 7.5

$$\frac{K_H}{K_D} = 7.5 \quad \therefore \frac{K_{100\% H_2O}}{K_{100\% D_2O}} = \frac{1}{7.5} = 0.13 \quad (41)$$

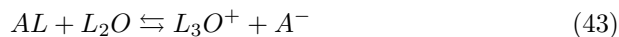
And we know that...

$$\begin{aligned} \frac{K_{100\% D_2O}}{K_{100\% H_2O}} &= \frac{(\phi_{L_3O})^3 \cdot \phi_{OL}}{(\phi_{L_2O})^2} \\ 0.13 &= \frac{(0.69)^3 \cdot \phi_{OL}}{(1)^2} \\ \phi_{OL} &= 0.41 \end{aligned} \quad (42)$$

Equation 42 shows that ϕ_{OL} is less than 1, which indicates that the force constant for the O—L bond in LO^- is weaker than in water. This is a surprising result but can be explained by the strong hydrogen-bonding network around a hydroxide ion. It will be hydrogen bonded most of the time to three water molecules, which will reduce electron density at the oxygen.

6.2.2 Solvent isotope Effects on Weak Acid/base Equilibria

For the equilibrium for a weak base...



The solvent isotope effect is defined by...

$$\frac{K_{100\% D_2O}}{K_{100\% H_2O}} = \frac{(\phi_{L_3O^+})^3 \cdot \phi_{A^-}}{\phi_{AL} \cdot (\phi_{L_2O})^2} \quad (44)$$

The fractionation factor for L_2O (water or D_2O) is 1 by definition.. As in the case of hydronium ion, the fractionation factor for the protonated acid (ϕ_{AL}) will be less than 1. But it will be closer to 1 than in the case of water. As in the case of hydroxide ion the fractionation factor for the deprotonated acid (ϕ_{A^-}) will be less than 1, but since it will be solvated less effectively than hydroxide, it will be closer to 1. For simplicity, we will assume that the ϕ_{AL} and ϕ_{A^-} will be similar in magnitude and will cancel each other in Equation 44.

Using this simplification we can see Equation 44 become...

$$\frac{K_{100\% D_2O}}{K_{100\% H_2O}} = (\phi_{L_3O^+})^3 = (0.69)^3 = 0.33 \quad (45)$$

So the kinetic isotope effect for acid dissociation of a weak acid in water compared to D_2O is...

$$\frac{K_H}{K_D} = \frac{1}{0.33} = 3 \quad (46)$$

This value is in agreement with experimental data.

Note: You may noticed that the K_H/K_D for water acid dissociation is 7.5. The difference is mostly due to the fact that ϕ_{AL} and ϕ_{A^-} do not in fact cancel exactly. The value of K_H/K_D we obtained for the fictional acid above assumes that $\phi_{AL} \approx \phi_{A^-}$. This is not always the case (see water example) and so we will have a range of possible values for K_H/K_D but they will be close to 3.

So we see that acids are three times stronger in H_2O compared to D_2O . This corresponds to a pK_a difference of...

$$\log \frac{K_H}{K_D} = \log K_H - \log K_D = pK_D - pK_H = \log(3) = 0.47 \quad (47)$$

So the pK_a of an acid is about 0.5 pK_a units lower for acids in water compared to acids in D_2O . So acids based on oxygens (e.g. H_2SO_4 , CH_3COOH , CH_3OH , etc...) are stronger by about a half pK_a unit in water compared to in D_2O .